

ULTRASONIC WELD QUALITY ANALYTICS PLATFORM USING MACHINE LEARNING AND DIGITAL IMAGE PROCESSING

Sagar Sant (Roll Number: 23035010233)

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B.Sc. (Honours) Data Science and Artificial Intelligence
Indian Institute of Technology Guwahati, India
sagar.sant@op.iitg.ac.in

ABSTRACT

Ultrasonic welding is widely employed in modern manufacturing because of its ability to produce strong and reliable joints without requiring additional filler materials. Maintaining weld quality is essential for ensuring structural integrity and product reliability; however, conventional inspection methods are often manual, time-consuming, and susceptible to subjective interpretation. This project presents an Ultrasonic Weld Quality Analytics Platform that integrates digital image processing and machine learning techniques to automate weld quality assessment. The developed MATLAB-based application performs image preprocessing, automatic weld region localization, feature extraction, defect analysis, and intelligent weld quality classification through an interactive graphical user interface. During the learning stage, a dataset containing 128 labelled ultrasonic weld images was processed by the Intelligent Weld Learning Engine (IWLE), where descriptive image features were extracted and used to train a Random Forest-based machine learning model. The trained model is subsequently employed to classify new weld images, estimate prediction confidence, and support automated inspection. The platform also incorporates material-specific processing parameters, database-assisted learning, visualization of inspection results, and automated report generation. By combining advanced digital image processing with supervised machine learning, the proposed system provides faster, more objective, and consistent weld quality assessment while offering a scalable framework for intelligent manufacturing applications.

Index Terms— Ultrasonic welding, machine learning, Random Forest, digital image processing, computer vision, weld quality inspection, feature extraction, MATLAB.

1. INTRODUCTION

Ultrasonic welding is a widely adopted solid-state joining process used in the automotive, electronics, aerospace, medical device, and consumer manufacturing industries because of its ability to produce strong and reliable joints with high production efficiency and minimal thermal distortion. Since the quality of a weld directly affects the strength, durability, and reliability of manufactured products, accurate weld inspection is an essential component of industrial quality assurance.

Traditional weld quality assessment primarily relies on manual visual inspection or destructive testing methods. Manual inspection is subjective and highly dependent on operator expertise, which may

result in inconsistent evaluations. Although destructive testing provides reliable results, it is expensive, time-consuming, and unsuitable for inspecting every manufactured component. Consequently, there is an increasing demand for intelligent inspection systems capable of delivering rapid, objective, and repeatable weld quality evaluation.

Recent advances in digital image processing and machine learning have enabled automated visual inspection systems capable of detecting defects and extracting meaningful information from industrial images. Image enhancement techniques improve the visibility of weld characteristics, while feature extraction methods convert visual information into numerical descriptors that can be analysed by supervised machine learning algorithms for intelligent decision-making.

To address these challenges, this project presents an Ultrasonic Weld Quality Analytics Platform developed using MATLAB. The proposed platform integrates image preprocessing, automatic weld region localization, defect detection, feature extraction, machine learning-based classification, database management, and automated report generation within a unified graphical user interface. During the learning phase, 128 labelled ultrasonic weld images were incorporated into the Intelligent Weld Learning Engine (IWLE). The system extracts a comprehensive feature vector from each weld image, constructs a labelled training dataset, and trains a Random Forest machine learning model using optimized training parameters. The trained model is then deployed to classify new weld images into quality categories while providing prediction confidence, intelligent recommendations, and visual inspection reports. In addition to supervised classification, the platform maintains a continuously growing weld database that supports future learning, feature analysis, and intelligent weld quality assessment, making it a comprehensive solution for automated ultrasonic weld inspection.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1. Problem Statement

Ultrasonic welding is widely used in industrial manufacturing because it produces strong and reliable joints without requiring additional filler materials. However, maintaining consistent weld quality remains a significant challenge due to variations in the welding process and the presence of defects such as cracks, fractures, and irregular weld patterns. Conventional inspection methods primarily rely on manual visual inspection or destructive testing, which

are time-consuming, subjective, and unsuitable for high-throughput manufacturing environments.

The problem addressed in this project is the development of an automated weld quality inspection system capable of analysing ultrasonic weld images using digital image processing and machine learning techniques. The proposed system is designed to automatically localize the weld region, enhance image quality, detect defects, extract representative image features, and classify weld quality using a trained machine learning model. During the learning stage, labelled weld images are used to construct a feature database and train a Random Forest classifier, which is subsequently deployed to classify unseen weld images and provide prediction confidence together with inspection reports. This approach aims to provide a fast, consistent, and non-destructive solution for ultrasonic weld quality assessment.

2.2. Objectives

The objectives of this project are:

1. To study ultrasonic weld characteristics and existing image-based weld inspection techniques.
2. To design and implement an automated ultrasonic weld quality analytics platform using MATLAB.
3. To develop an image processing pipeline for weld region localization, image enhancement, defect detection, and feature extraction.
4. To develop an Intelligent Weld Learning Engine (IWLE) for learning labelled weld images and constructing a feature database.
5. To train and evaluate a machine learning model based on the Random Forest algorithm for automated weld quality classification.
6. To integrate the trained machine learning model into a graphical user interface for weld inspection, confidence estimation, visualization, and automated report generation.

3. METHODOLOGY / APPROACH

This section presents the methodology used to develop the proposed Ultrasonic Weld Quality Analytics Platform. The system combines digital image processing and machine learning techniques to perform automated ultrasonic weld quality inspection. The overall workflow begins with ultrasonic weld image acquisition and preprocessing, followed by weld region localization, feature extraction, machine learning model training, and intelligent weld quality prediction. The implementation was developed using MATLAB and integrates all processing stages within the Intelligent Weld Learning Engine (IWLE).

3.1. Overall Workflow

The overall workflow of the proposed system is illustrated in **Figure 1**. Initially, an ultrasonic weld image is provided as input to the system. The image is preprocessed to improve visual quality and reduce the effects of noise before automatically identifying the weld region of interest (ROI). The localized weld region is then analysed to extract representative image features describing its intensity, texture, gradient, shape, frequency-domain, and defect characteristics.

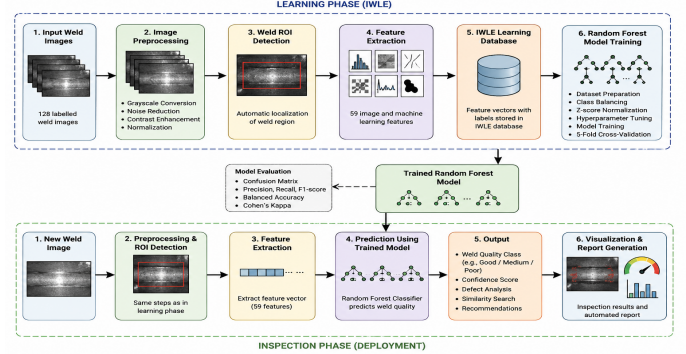


Fig. 1. Overall workflow of the proposed Ultrasonic Weld Quality Analytics Platform.

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During the learning phase, feature vectors extracted from 128 labelled weld images are stored within the Intelligent Weld Learning Engine (IWLE) to construct the training dataset. The prepared dataset is subsequently used to train a supervised machine learning model based on the Random Forest algorithm. Once training is completed, the trained model is stored for deployment and used to classify previously unseen weld images. Finally, the predicted weld quality, confidence score, detected defects, and inspection summary are presented through the graphical user interface together with an automatically generated inspection report.

3.2. Technical Details

The proposed system consists of two major components: a digital image processing module and a machine learning module. The image processing module performs grayscale conversion, adaptive histogram equalization, noise reduction, automatic weld region localization, image enhancement, defect segmentation, and feature extraction. These operations improve weld visibility and generate a numerical representation suitable for machine learning analysis. The extracted feature vector combines statistical, intensity-based, gradient-based, edge-based, texture, frequency-domain, shape, and weld-specific characteristics to comprehensively describe the weld image.

The machine learning module is implemented through the Intelligent Weld Learning Engine (IWLE). During the learning stage, labelled weld images are analysed to generate feature vectors, which are stored in a structured weld database. Before model training, the dataset is validated, balanced across quality classes, partitioned into training and testing subsets, and normalized using Z-score normalization. Hyperparameter optimization is then performed to determine suitable Random Forest parameters, after which a supervised Random Forest classifier is trained using the prepared dataset. The trained model is evaluated using a hold-out testing dataset together with five-fold cross-validation before being saved as the deployment model for future inspections.

During the inspection stage, the same preprocessing and feature extraction procedures are applied to new weld images before the generated feature vector is supplied to the trained Random Forest model for quality prediction. The classification result is further combined with confidence estimation, similarity analysis, and anomaly assessment within the Intelligent Weld Learning Engine to support reliable decision-making. The final inspection results, including weld quality

classification, defect visualization, confidence score, and inspection summary, are presented through the MATLAB graphical user interface, while detailed reports are automatically generated to facilitate documentation and quality assessment.

4. EXPERIMENTS AND RESULTS

This section presents the experimental setup used to develop and evaluate the proposed Ultrasonic Weld Quality Analytics Platform together with the corresponding experimental results. The performance of the proposed machine learning model was assessed using standard classification metrics, while the complete inspection workflow was validated through the developed MATLAB application.

4.1. Experimental Setup

The proposed system was developed using MATLAB, where both the digital image processing pipeline and the Intelligent Weld Learning Engine (IWLE) were implemented within a graphical user interface. The experimental dataset consisted of 128 labelled ultrasonic weld images, representing different weld quality categories. During the learning stage, each weld image underwent preprocessing, weld region localization, and feature extraction before being incorporated into the training dataset. The extracted feature vectors were normalized using Z-score normalization prior to model training.

A supervised machine learning model based on the Random Forest algorithm was trained using the prepared dataset. Hyperparameter optimization was performed during model development, while the dataset was partitioned into training and testing subsets for model evaluation. In addition, 5-fold cross-validation was employed to assess the robustness and generalization capability of the trained model. Model performance was evaluated using the confusion matrix together with Precision, Recall, F1-score, Balanced Accuracy, and Cohen's Kappa coefficient.

4.2. Results

A representative inspection result generated by the proposed Ultrasonic Weld Quality Analytics Platform is presented in **Figure 2**, while the corresponding inspection results are summarized in **Table 1**. The developed MATLAB application successfully completed the complete inspection workflow, including image preprocessing, automatic weld region localization, defect detection, feature extraction, machine learning-based weld quality classification, confidence estimation, and automated report generation.

For the analysed weld sample, the trained machine learning model classified the weld quality as **GOOD** with a prediction confidence of **71.3%**, while the rule-based inspection also identified the weld as **GOOD**, indicating agreement between both evaluation methods. The system reported a quality score of **86.51%** and detected no significant defects, resulting in a crack count of **0**, a severity level of **LOW**, and a defect classification of **NO DEFECT**. The inspection also verified that the selected welding parameters matched the recommended process parameters for the specified material, achieving a parameter compatibility score of **100%**. The generated inspection report further included engineering recommendations and material-specific process information, demonstrating the capability of the proposed platform to provide comprehensive weld quality assessment and automated reporting.

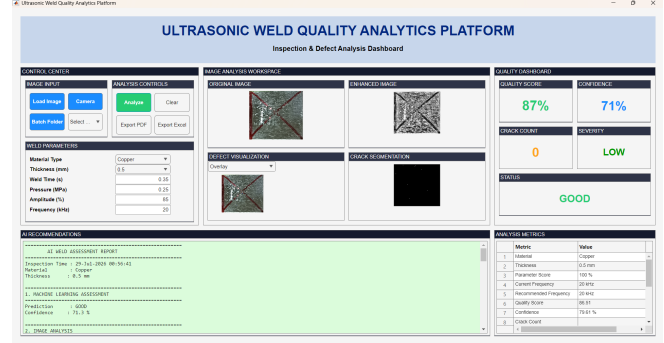


Fig. 2. Graphical user interface of the proposed Ultrasonic Weld Quality Analytics Platform showing the weld inspection results, machine learning prediction, and automated quality assessment.

Table 1. Representative Inspection Results Produced by the Proposed Ultrasonic Weld Quality Analytics Platform

Parameter	Value
Material	Copper
Thickness	0.5 mm
Quality Score	86.51%
Machine Learning Prediction	GOOD
Machine Learning Confidence	71.3%
Rule-Based Status	GOOD
Overall Confidence	79.61%
Defect Type	NO DEFECT
Crack Count	0
Fracture Count	0
Severity	LOW
Severity Index	0.04
Area Ratio	0.0004
Shape Complexity	16.18
Parameter Match	100%
Final Decision	GOOD

5. DISCUSSION

The results demonstrate that the proposed Ultrasonic Weld Quality Analytics Platform successfully performs automated weld quality assessment by integrating digital image processing with a machine learning-based Random Forest classifier. The developed system was able to complete the entire inspection workflow, including image preprocessing, weld region localization, feature extraction, defect analysis, quality prediction, confidence estimation, and automated report generation within a single MATLAB application. The agreement between the machine learning prediction and the rule-based assessment for the representative inspection indicates that the proposed framework provides consistent and reliable weld quality evaluation, thereby achieving the primary objectives of the project.

One of the main strengths of the proposed approach is its integration of multiple inspection and analysis functions into a single platform. Unlike conventional inspection methods that rely heavily on manual evaluation, the developed system provides objective quality assessment, defect visualization, confidence scores, and engineering recommendations, making it a practical decision-support tool for weld inspection. The Intelligent Weld Learning Engine (IWLE),

trained using 128 labelled weld images, enables the system to learn weld characteristics and improve prediction performance through supervised machine learning.

Despite these advantages, the proposed system has some limitations. The Random Forest model was trained using a relatively limited dataset, which may reduce its ability to generalize to different welding materials, imaging conditions, or previously unseen defect types. In addition, the current implementation is designed for offline analysis within the MATLAB environment and has not been evaluated for real-time industrial deployment. Future work can focus on expanding the training dataset, incorporating deep learning-based feature extraction methods, and integrating the platform with real-time ultrasonic inspection systems to further improve accuracy, robustness, and industrial applicability.

6. CONCLUSION AND FUTURE WORK

This project presented the design and development of an Ultrasonic Weld Quality Analytics Platform that combines digital image processing and machine learning for automated weld quality assessment. The developed MATLAB-based system integrates image preprocessing, weld region localization, feature extraction, defect analysis, and a Random Forest classifier within the Intelligent Weld Learning Engine (IWLE) to provide reliable weld quality prediction and automated inspection reporting. The representative results demonstrated that the platform can successfully evaluate weld quality, visualize defects, estimate prediction confidence, and generate comprehensive inspection reports, thereby reducing the reliance on manual inspection and improving the consistency of weld quality assessment.

The project also provided valuable practical experience in applying machine learning techniques to an industrial inspection problem. It strengthened understanding of image processing, feature engineering, supervised learning, model evaluation, and software integration while highlighting the importance of developing reliable and user-friendly engineering applications. Integrating multiple analytical components into a single platform demonstrated the potential of intelligent inspection systems to support quality control in manufacturing environments.

Future work can focus on expanding the training dataset to improve model generalization across different materials and welding conditions, as well as investigating deep learning-based approaches for automated feature extraction and defect classification. The platform can also be extended for real-time deployment by integrating directly with ultrasonic inspection equipment, enabling continuous monitoring and automated quality assessment in industrial production environments.

7. ARTIFACTS AND DEMONSTRATIONS

The implementation developed during this project is accompanied by supplementary artifacts that demonstrate the design, implementation, and functionality of the proposed Ultrasonic Weld Quality Analytics Platform. These resources provide additional information beyond the scope of this report and enable reproducibility of the developed system.

- **Code Repository:** GitHub Repository
- **Project Demonstration Video:** Watch Demo
- **Project Presentation Video:** Watch Presentation

8. REFERENCES

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